

Automorphisms of field extensions

Recall:

k : finite field

$$(1) |k| = p^d \text{ for a prime } p$$

$$(=) \mathbb{F}_p \leq k : \text{field extension}$$

$$(2) k^\times \text{ is a cyclic group of order } p^d - 1$$

$$(3) \exists \text{ a field automorphism}$$

$$\begin{array}{ccc} \text{Fr} : k & \longrightarrow & k \\ \alpha & \longmapsto & \alpha^p \end{array}$$

$$\text{Key: } (\alpha + \beta)^p = \alpha^p + \beta^p \quad]$$

Properties of Fr

(1)

$$\text{Fr}^d = \text{id}$$

$$\Leftrightarrow \begin{matrix} \mathbb{Z}/d\mathbb{Z} \\ \cong \\ \langle \text{Fr} \rangle \end{matrix} \hookrightarrow k$$

In fact,
Fr has order
d

$$\Leftrightarrow \alpha^{p^d} = \alpha \quad \forall \alpha \in k$$

(2) $k_r = \{ \alpha \in k : \text{Fr}^r(\alpha) = \alpha \}$

is a subfield of k
of order $p^{\text{ord}(\text{Fr}^d)}$

(3) (1) & (2) together can be interpreted as follows

$$\begin{matrix} \mathbb{Z}/d\mathbb{Z} \\ \cong \\ G = \langle \text{Fr} \rangle \end{matrix} \hookrightarrow k$$

Vl

$G_m \leftarrow$ unique subgroup, $G_m = \langle \text{Fr}^{d/m} \rangle$
of order $m \mid d$

$$k^{G_m} \stackrel{\text{def}}{=} \left\{ \alpha \in k : \sigma(\alpha) = \alpha \right\}$$

$$\forall \sigma \in G_m$$

$$= \left\{ \alpha \in k : \text{Fr}^{d/m}(\alpha) = \alpha \right\}$$

$$|k^{G_m}| = p^{d/m}$$

Example: $d = 12$

$$G = \langle \text{Fr} \rangle, |G| = 12$$

$$\langle \text{Fr}^4 \rangle$$

$$G_1 = \langle \text{Fr}^3 \rangle$$

$$G_2 = \langle \text{Fr}^6 \rangle$$

$$\mathbb{F}_{p^4} \leq \mathbb{F}_{p^{12}}^{\text{VI}}$$

$$\mathbb{F}_{p^6}$$

$$\langle \text{Fr}^2 \rangle = G_6$$

$$G_4 = \langle \text{Fr}^3 \rangle$$

$$\mathbb{F}_{p^2} = k^{G_6}$$

$$k^{G_4} = \mathbb{F}_{p^3}$$

$$G = G_{12}$$

$$\langle \text{Fr} \rangle$$

$$\mathbb{F}_p = k^G$$